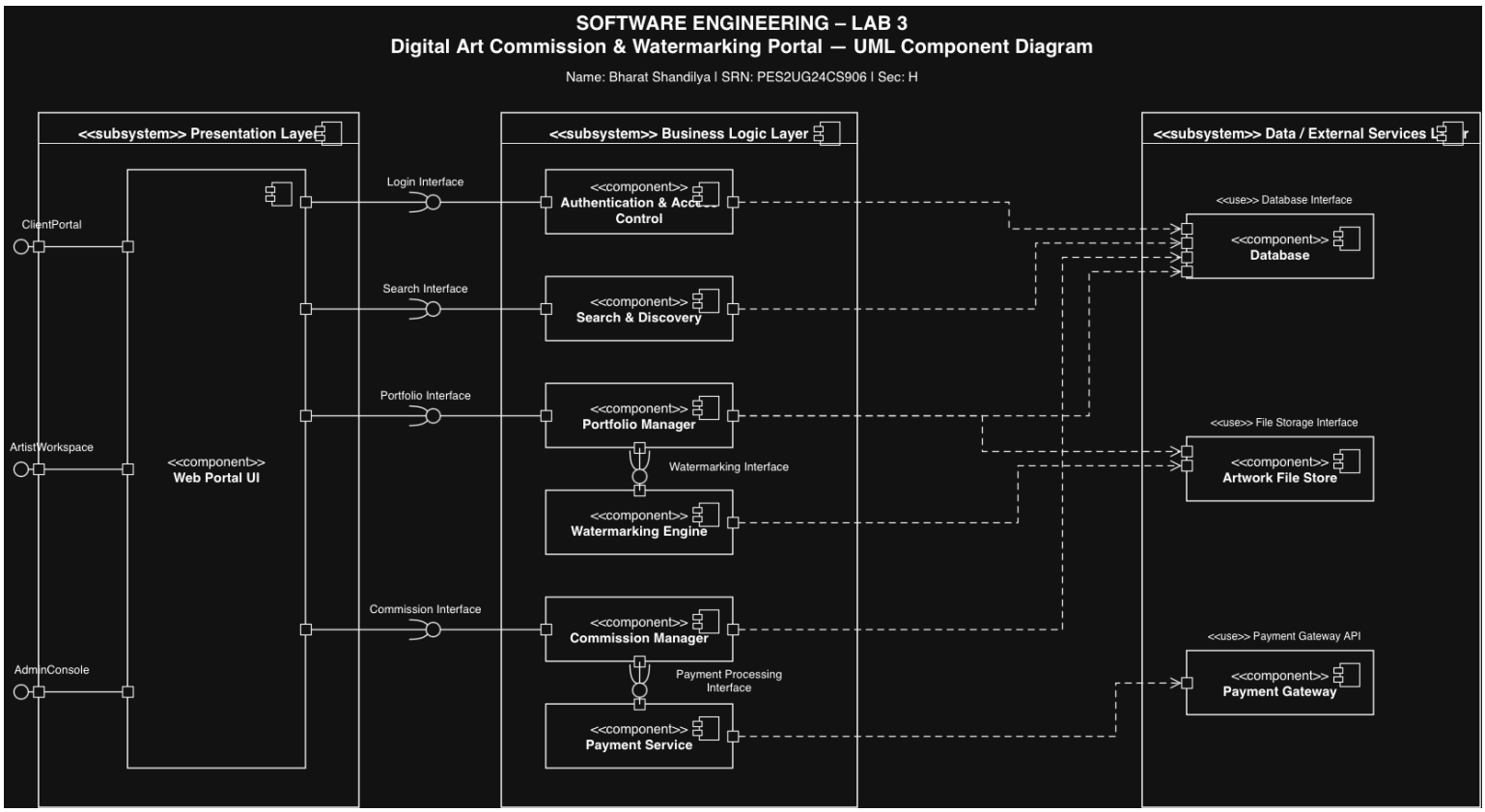
SOFTWARE ENGINEERING - LAB 3
Digital Art Commission & Watermarking Portal system

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Component Diagram:



Justification

Layered architecture is used for this system.
The system is divided into Presentation, Business Logic, and Data/External Services layer. The Web Portal UI is for clients, digital artists, and administrators. The business layer handles authentication, searching, portfolio management, commissions, watermarking, and payments. The data and external services layer manages the database, artwork file storage, and payment gateway.

Reason 1: Layered architecture is suitable because the system follows a clear workflow. Users log in, search and view artwork, manage portfolios, request commissions, apply watermarks, and make payments through separate layers.

Reason 2: It separates responsibilities between components. Authentication, search, portfolio, commission, watermarking, and payment functions can be developed and maintained independently. The scalability is very high.

Security: The Authentication & Access Control component controls user access and ensures that clients, artists, and administrators can access only the functions available to them.

Performance: Separating the major functions allows operations such as artwork search, commission management, watermarking, and payment processing to be handled independently.

Components:
Web Portal UI, Authentication & Access Control, Search & Discovery, Portfolio Manager, Watermarking Engine, Commission Manager, Payment Service, Database, Artwork File Store, and Payment Gateway.

Main Interfaces
Login Interface, Search Interface, Portfolio Interface, Watermarking Interface, Commission Interface, Payment Processing Interface, Database Interface, File Storage Interface, and Payment Gateway API.

Conclusion
Layered Architecture is suitable for the Digital Art Commission & Watermarking Portal because it provides a simple structure, clear separation of responsibilities, and easy communication between the user interface, business components, data storage, and external services.